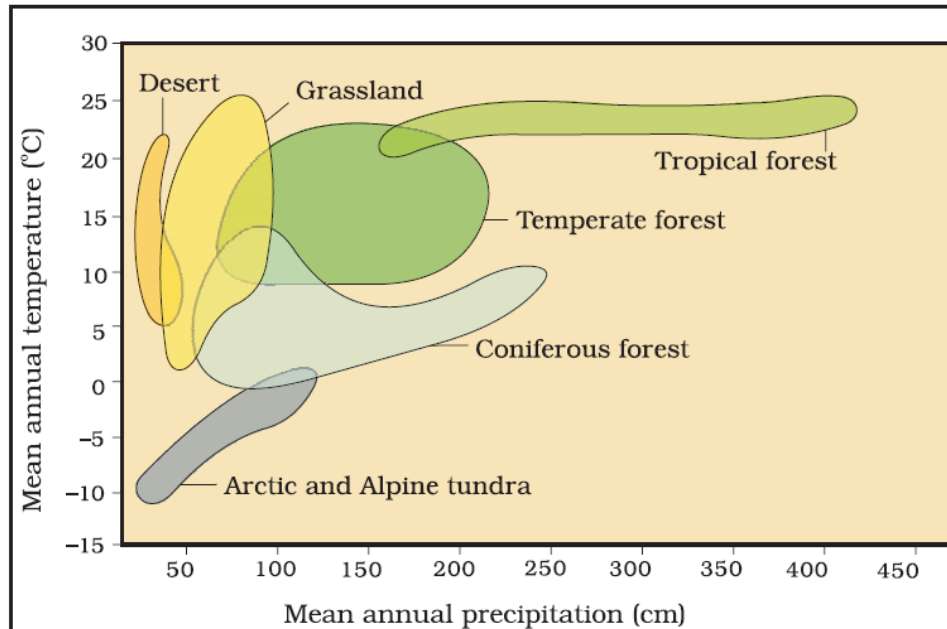


Organisms and Environment

- **Ecology** deals with interactions among different organisms and their environment.
- Organisms get adapted to their environment for their survival and reproduction.
- The rotation of the earth about its axis brings about changes in the environment, leading to different seasons. This leads to the formation of various biomes such as desert, grassland, etc.



Biome distribution with respect to annual temperature and precipitation

- **Major biomes of India** – (a) Tropical rain forest; (b) Tropical rain forest; (b) Deciduous forest; (c) Desert; (d) Sea coast



Tropical rain forest



Deciduous forest



Desert



Sea coast

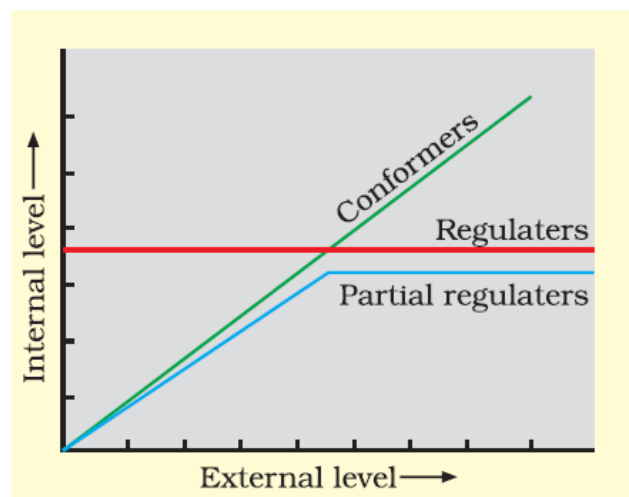
- Life not only exists in favourable habitats, but also in harsh and extreme conditions.
- The environment of an organism can be divided into: **Abiotic factors** **Biotic factors**

Abiotic Factors

- Some of the major abiotic factors that interact with the organisms are:
- Temperature** – It is the most relevant abiotic factor since all organisms require an optimum temperature for their metabolism and other body functions. Depending upon their ability to tolerate temperature range, organisms are of two types- **stenothermal** (restricted to a narrow range of temperature) and **eurythermal** (can tolerate a wide range of temperature).
- Water** – Water also is a major influencing factor. Life on earth is impossible without water as it forms the major constituent of all living cells. In oceans where quantity of water is not a limitation, the quality of water becomes one. Depending upon the ability to tolerate salinity (the salt concentration is measured as salinity in parts per thousand), organisms can be **stenohaline** (restricted to narrow range of salinity) and **euryhaline** (tolerant to wider range of salinity).
- Soil** – The nature and composition of soil differs from one place to another depending upon the climate, weathering process, and soil development method. The characteristic features such as soil composition, grain size, percolation, water holding capacity, etc. determine the native of the organisms it can support.
- Light** – The major source of light on earth is the Sun. Light is essential for plants to perform photosynthesis. Certain plants become adapted to perform photosynthesis under very low light since they are constantly overshadowed by tall trees. Many plants also depend on light for their flowering (**photoperiodism**). The availability of light on land is comparatively higher than that in water.

Responses to Abiotic Factors

- All organisms in order to sustain maximum functionality maintain a constant internal environment (**homeostasis**). An organism may adopt one of the following strategies for homeostasis:
- Regulate** – certain animals have the ability to maintain a constant temperature and a constant osmolality to keep up their homeostasis such animals are called as **regulators**. Mammals have a constant body temperature (37°C) irrespective of the outside temperature. In summers, to maintain the temperature, we sweat and in winters we shiver, which produces heat.
- Conform** – Animals and plants except mammals do not have a constant body temperature and their body temperature changes in accordance with the outside temperature. Such organisms are called **conformers**. Conformers have not evolved. They have not become regulators since regulation is energetically more expensive.
- Migrate** – The organism can move temporarily from stressful habitats to more hospitable areas and return once the period changes. Birds can migrate from cold regions to relatively warmer regions during winter and vice-versa during summers.
- Suspend** – Some organisms cease to be metabolically active during stressful period. They suspend all activity and enter a period of **dormancy**. For example – Spores in bacteria and lower plants; and **hibernation** (winter sleep) and **aestivation** (summer sleep) in animals similarly, zooplankton enter **diapause**, a stage of suspended development during unfavourable conditions.



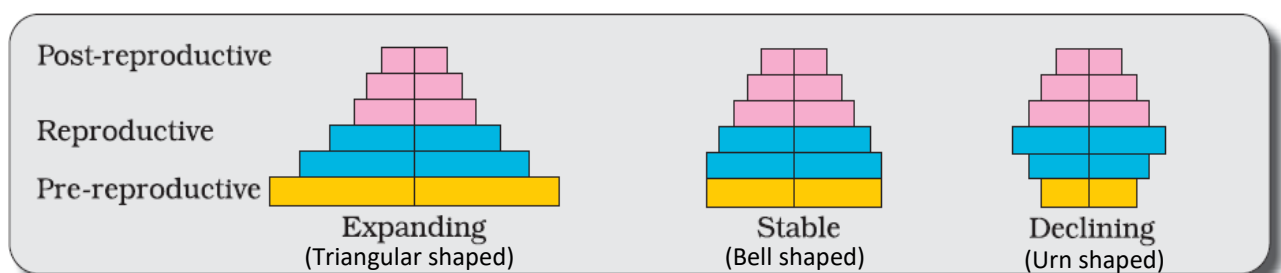
Diagrammatic representation of organismic response

Adaptations

- Adaptations are certain characteristics that organisms develop in order to survive and reproduce better in their habitat.
- These adaptations can be physiological, behavioural, or morphological.
- Some of the adaptations are:
 - **Desert plants** have thick cuticle on their leaf surface and stomata arranged in deep pits to reduce water loss. Their special photosynthetic pathway CAM (Crassulacean acid metabolism) enables their stomata to remain closed during day time. Their leaves are reduced to spines and photosynthesis is carried out by flattened stems.
 - **Animals of colder regions** have shorter limbs and ears to minimise heat loss (**Allen's rule**) and the body is covered by thick fur to reduce the heat loss. In the polar seas aquatic mammals like seals have a thick layer of fat (blubber) below their skin that acts as an insulator to minimise heat loss.
 - **People living in high altitudes** (>3,500m Rohtang Pass near Manali and Mansarovar, in China occupied Tibet) have high RBC production and increased breathing rates than the people living in plains. If you had ever been to any high altitude place you must have experienced what is called **altitude sickness** – its symptoms include nausea, fatigue and heart palpitation. This is because in the low atmospheric pressure of high altitudes, the body does not get enough oxygen. But, gradually you get acclimatized and stop experiencing altitude sickness. The body compensates low oxygen availability by increasing red blood cell production, decreasing the binding capacity of hemoglobin and by increasing breathing rate.
 - Some **desert animals** are capable of burrowing in order to escape the heat. In addition, some desert animals such as kangaroo rat are able to meet their water requirement through internal fat oxidation. They also have ability to concentrate their urine. Desert lizards bask in the sun and absorb heat when their body temperature drops below the comfort zone, but move into shade when the ambient temperature starts increasing. Some species are capable of burrowing into the soil to hide and escape from the above ground heat.

Population:

- It is a group of similar individuals living in a geographical area, sharing similar resources, and capable of interbreeding.
- Population has certain attributes, which individual organisms do not possess:
 - **Birth rate** – the number of live births per one thousand people per year (per capita births)
 - **Death rate** – the number of deaths per one thousand people per year (per capita deaths)
 - **Sex ratio** – Ratio of number of males to females in a population
- **Age distribution**
 - A population can be composed of individuals of different age groups.
 - Age distribution plot for a given population is given by the **age pyramid**.
 - The structure of the age pyramid determines the growth status of the population, whether it is growing, stable, or declining.



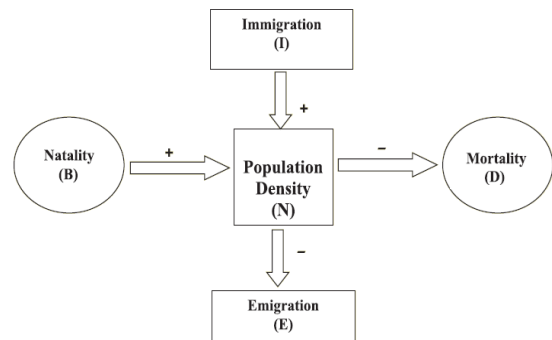
Representation of age pyramids for human population

- Population size, more technically, is referred to as **population density** (N), which indicates the number of individuals inhabiting a particular **niche**.
- If the population is huge, then relative density is measured instead of absolute density whose measurement is time-consuming.

Population Growth

- The size of a population is an ever-changing aspect since it depends upon availability of food, predation, weather conditions, etc.
- This gives us an idea whether a certain population is growing or declining.
- Some of the reasons for the increase or decrease in population:

- Natality** (B) – Number of births during a given period in the given population
- Mortality** (D) – Number of deaths during a given period in the given population
- Immigration** (I) – Number of individuals of the same species who have come into the habitat from elsewhere during a given period
- Emigration** (E) – Number of individuals of the same species who have left the habitat and gone elsewhere during a given period



- If N is the population at time t, then its density at t + 1 is

$$N_{t+1} = N_t + [(B + I) - (D + E)]$$

Growth Models

- Exponential Growth** – When the resources are unlimited, population tends to grow in an exponential pattern.
 - If the population size is N and the birth and death rates are b and d respectively, then increase or decrease in N at t (time period) is given by,

$$dN / dt = (b - d) \times N$$

If $(b - d) = r$, then

$$dN/dt = rN$$

Where, “r” is the “intrinsic rate of natural increase”.

Or,

The integral form of the exponential growth equation as

$$N_t = N_0 e^{rt}$$

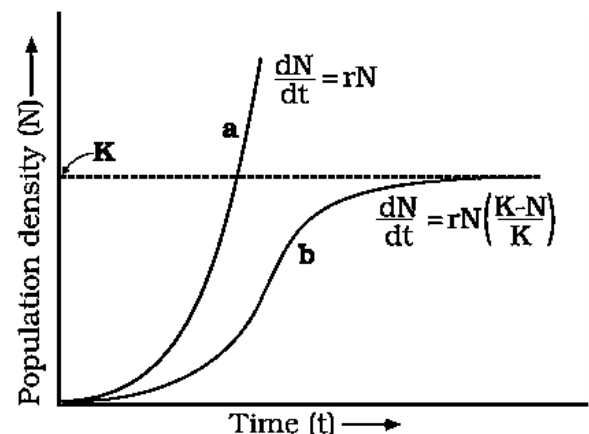
Where,

N_t – Population density at time t

N_0 – Population density at time 0

r – Intrinsic rate of natural increase

e – Base of natural logarithms (2.71828)



Population growth curve - **a**. When responses are not limiting the growth, plot is exponential, **b**. when responses are limiting the growth, plot is logistic, and **K** is carrying capacity

- Logistic growth** – When the resources are limited leading to competition between individuals and survival of the fittest, the population tends to grow in a logistic manner.
 - In this kind of growth, there is an initial lag phase followed by acceleration or deceleration phases and finally asymptote, when it reaches its carrying capacity (K).
 - When N in relation to t is plotted, it results in a sigmoid curve called the **Verhulst – Pearl Logistic growth** given by,

$$\frac{dN}{dt} = rN \left(\frac{K - N}{K} \right)$$

Where,

N – Population density at time t

r – Intrinsic rate of natural increase

K – Carrying capacity

Life History Variations

- Populations tend to increase their reproductive fitness in order to survive better. This is known as Darwinian fitness (high r value).
- Some of the trends they follow in course of achieving this:
 - Some organisms breed only once in their lifetime. Example – Salmon fish, Bamboo
 - Some breed many times. Example – most Birds, Mammals
 - Some produce a large number of small-sized offsprings. Example – Oysters, Pelagic fishes
 - Some produce small number of large-sized offsprings. Example - Birds, Mammals

Population Interactions

- A natural habitat consists of many organisms living together and these organisms communicate and interact with each other. For example, plants depend on insects for pollination.
- Interspecific interactions are interactions between two different species of organisms. They can be either beneficial or harmful to one or both partners.

Interspecific interactions

- **Predation** – It is beneficial to the predator while the prey is harmed.
 - It acts as a means of transfer of energy to the next higher trophic level and of maintaining balance in the ecosystem.
 - For plants, herbivores are predators and some plants produce secondary metabolites, thorns, or poisonous chemicals to ward off predators.
 - Similarly, animals also camouflage themselves to protect themselves from predators. Some preys are poisonous or distasteful (Monarch butterfly is highly distasteful because of a special chemical it acquires during its caterpillar stage by feeding on poisonous weeds) so as to avoid predators.
- **Competition** – It occurs only in closely related species wherein they share the same type of habitat and food resources.
 - However, for competition to take place resources need not be always scarce and competition does not necessarily take place between same species.
 - In competition, the fitness of one species is significantly lower in presence of another species and survival of fittest ultimately takes place.
 - **Gause's Competitive Exclusion Principle** states that two closely related species competing for the same resource cannot co-exist indefinitely and the competitively inferior will be eliminated eventually.
 - Moreover, some species may develop mechanisms to facilitate their co-existence.
- **Parasitism** – In this interaction, one of the partners is benefited because it resides outside or inside the body of the host and gets free accommodation and food while the host is affected due to loss of nutrients.
 - Parasites in nature have developed a wide variety of adaptations such as hooks and suckers for attachment, loss of digestive system, high reproducing capacity, etc.

- Parasites can live either outside (**ectoparasites**: - lice on humans and ticks on dogs, many marine fish are infested with ectoparasitic copepods, *Cuscuta*, a parasitic plant that is commonly found growing on hedge plants, has lost its chlorophyll and leaves in the course of evolution. It derives its nutrition from the host plant which it parasitises) or inside (**endoparasites**: - Plasmodium, liver fluke, tape worm, round worm) the body of the host organisms.
- Brood parasitism (Nest parasitism or Social parasitism) is seen in birds in which the parasitic bird lays its egg in the nest of the unassuming host bird, which takes care of them until they hatch. For example, Koel lays its eggs in the nest of the crow.
- **Commensalism** – In this interaction, one of the partners is benefited while the other is neither benefited nor harmed. For example,
 - An orchid growing as an epiphyte on the mango tree, and barnacles growing on the back of a whale benefit while neither the mango tree nor the whale derives any apparent benefits. The cattle egret and grazing cattle in close association.
 - The egrets always forage close to where the cattle are grazing because the cattle, as they move, stir up and flush out from the vegetation insects that otherwise might be difficult for the egrets to find and catch.
 - Interaction between sea anemone that has stinging tentacles and the clown fish that lives among them. The fish gets protection from predators which stay away from the stinging tentacles. The anemone does not appear to derive any benefit by hosting the clown fish.
- **Mutualism or symbiosis** – In this interaction, both the partners are benefited. For example,
 - Lichens, interaction of algae and fungi, where both are benefited. The fungi give support to the algae while the algae prepare the food for the fungi.
 - The *mycorrhizae* are associations between fungi and the roots of higher plants. The fungi help the plant in absorption of essential nutrients from the soil while the plant in turn provides the fungi with energy yielding carbohydrates.
 - In many species of fig trees, there is a tight one to one relationship with the pollinator species of wasp. The female wasp uses the fruit not only as an oviposition (egg- laying) site but uses the developing seeds within the fruit for nourishing its larvae, the wasp pollinates the fig inflorescence while searching for suitable egg-laying sites. In return for the favour of pollination the fig offers the wasp some of its developing seeds, as food for the developing wasp larvae.



(a)



(b)

Mutual relationship between fig tree and wasp: (a) Fig flower is pollinated by wasp: (b) Wasp laying eggs in a fig fruit

- The Mediterranean orchid *Ophrys* employs 'sexual deceit' to get pollination done by a species of bee. One petal of its flower bears an uncanny resemblance to the female of the bee in size, colour and markings. The male bee is attracted to what it perceives as a female, 'pseudocopulates' with the flower, and during that process is dusted with pollen from the flower. When this same bee 'pseudocopulates' with another flower, it transfers pollen to it and thus, pollinates the flower. This is also an example for co-evolution.



Figure: Showing bee a pollinator on orchid flower

Table: Population Interactions		
Species A	Species B	Name of Interaction
+	+	Mutualism
-	-	Competition
+	-	Predation
+	-	Parasitism
+	0	Commensalism
-	0	Amensalism

SUMMARY

As a branch of biology, Ecology is the study of the relationships of living organisms with the abiotic (physico-chemical factors) and biotic components (other species) of their environment. It is concerned with four levels of biological organisation-organisms, populations, communities and biomes.

Temperature, light, water and soil are the most important physical factors of the environment to which the organisms are adapted in various ways. Maintenance of a constant internal environment (*homeostasis*) by the organisms contributes to optimal performance, but only some organisms (regulators) are capable of homeostasis in the face of changing external environment. Others either partially regulate their internal environment or simply conform. A few other species have evolved adaptations to avoid unfavourable conditions in space (migration) or in time (aestivation, hibernation, and diapause).

Evolutionary changes through natural selection take place at population level and hence, population ecology is an important area of ecology. A population is a group of individuals of a given species sharing or competing for similar resources in a defined geographical area. Populations have attributes that individual organisms do not- birth rates and death rates, sex ratio and age distribution. The proportion of different age groups of males and females in a population is often presented graphically as age pyramid; its shape indicates whether a population is stationary, growing or declining.

Ecological effects of any factors on a population are generally reflected in its size (population density), which may be expressed in different ways (numbers, biomass, per cent cover, etc.,) depending on the species.

Populations grow through births and immigration and decline through deaths and emigration. When resources are unlimited, the growth is usually exponential but when resources become progressively limiting, the growth pattern turns logistic. In either case, growth is ultimately limited by the carrying capacity of the environment. The intrinsic rate of natural increase (r) is a measure of the inherent potential of a population to grow.

In nature populations of different species in a habitat do not live in isolation but interact in many ways. Depending on the outcome, these interactions between two species are classified as competition (both species suffer), predation and parasitism (one benefits and the other suffers), commensalism (one benefits and the other is unaffected), amensalism (one is harmed, other unaffected) and mutualism (both species benefit). Predation is a very important process through which trophic energy transfer is facilitated and some predators help in controlling their prey populations. Plants have evolved diverse morphological and chemical defenses against herbivory. In competition, it is presumed that the superior competitor eliminates the inferior one (the Competitive Exclusion Principle), but many closely related species have evolved various mechanisms which facilitate their co-existence. Some of the most fascinating cases of mutualism in nature are seen in plant-pollinator interactions.